

Mini-Batch Clustering Analysis: Scalable K-Means and Fuzzy C-Means

**Comparative Study on High-Dimensional
and Large-Scale Tabular Data**

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1. Introduction

In the era of modern data science, clustering remains a cornerstone of exploratory data analysis, pattern recognition, and data compression. However, as datasets grow in both cardinality (number of samples) and dimensionality (number of features), traditional clustering algorithms face significant computational barriers. The most prominent example is Lloyd’s algorithm for K-Means, which requires a full pass over the dataset at every iteration, leading to $O(nkd)$ complexity.

This study explores **Mini-Batch variants** of two fundamental clustering paradigms: the “hard” boundary approach of **K-Means** and the “soft” membership approach of **Fuzzy C-Means (FCM)**. Mini-batch algorithms leverage stochastic approximation by updating model parameters using random subsets of data. This approach is particularly judicious when the dataset is too large to fit in memory or when rapid convergence is preferred over exact global optimization.

We evaluate these algorithms across two distinct domains:

1. **Tabular Data:** Using the *Covertypes* dataset (581k samples), representing a high-cardinality classification task.
2. **Computer Vision Data:** Using *Fashion-MNIST* (70k images), representing a high-dimensional feature space where traditional distance metrics behave differently.

The objective is to quantify the tradeoff between computational efficiency (runtime) and clustering quality (Silhouette, V-Measure, ARI), providing a rigorous assessment of when mini-batch stochasticity is preferable to full-batch exactitude.

2. Theoretical Framework

2.1. K-Means and the Mini-Batch Optimization

K-Means objective is to minimize the total within-cluster variance (Inertia):

$$J(X, \mu) = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$

The standard algorithm alternates between an **Assignment Step** and an **Update Step**. The bottleneck lies in the assignment $x \rightarrow \operatorname{argmin}_i \|x - \mu_i\|^2$, which must be computed for all n points.

Mini-Batch K-Means (Sculley, 2010) relaxes this constraint by using a small batch B of size b . For each $x \in B$, the algorithm finds the nearest center μ_i and updates that center using a learning rate η that decreases as more points are assigned to that cluster:

$$\mu_i \rightarrow (1 - \eta)\mu_i + \eta x$$

This stochastic gradient descent-like update allows the algorithm to converge much faster, often reaching an “acceptable” solution before having seen the entire dataset even once.

2.2. Fuzzy C-Means (FCM)

FCM introduces the concept of fuzzy sets to clustering. Each point x_j belongs to all clusters with varying degrees of membership $u_{ij} \in [0, 1]$. The fuzziness is controlled by the parameter m :

$$J_m = \sum_{i=1}^k \sum_{j=1}^n (u_{ij})^m \|x_j - \mu_i\|^2$$

When $m = 1$, FCM reduces to K-Means. Typically $m = 2$ is used to balance the “influence” of distant points. The fuzzy approach is particularly useful in datasets where clusters are not well-separated or where a point’s shared identity between two classes is semantically meaningful.

Mini-Batch FCM is an adaptation where only the points in the current batch contribute to the update of the membership matrix U and the centroids μ . This drastically reduces the memory footprint of the $n \times k$ membership matrix, which would otherwise be $O(nk)$.

3. Experimental Data and Preprocessing

3.1. Covertypes: Large-Scale Tabular Data

The **Covertypes** dataset (UCI) describes forest cover types from cartographic variables.

- **Cardinality:** 581,012 samples.
- **Dimensionality:** 54 features (Elevation, Aspect, Slope, Distances, etc.).
- **Preprocessing:** We apply **StandardScaler** to ensure all features contribute equally to the Euclidean distance calculation. This is critical because features like “Elevation” (thousands of meters) would otherwise dominate binary features like “Soil_Type” (0 or 1).
- **Subsampling:** 15,000 samples are used for the main evaluation to allow for multiple repetitions (10 seeds) while maintaining statistical significance.

3.2. Fashion-MNIST: High-Dimensional Imagery

Fashion-MNIST consists of 28×28 grayscale images.

- **Feature Space:** 784 dimensions (pixels).
- **Complexity:** Direct clustering in 784D is susceptible to the “Curse of Dimensionality,” where distances between all pairs of points become roughly equal.
- **Dimensionality Reduction:** We apply **Principal Component Analysis (PCA)** to reduce the space to 50 dimensions. This preserves 84% of the variance while significantly accelerating the distance computations ($\|x - \mu\|^2$) and improving the signal-to-noise ratio.

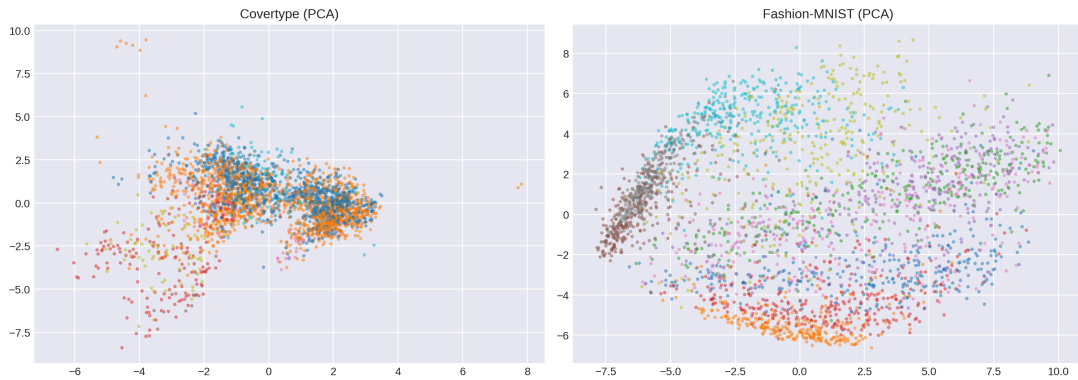


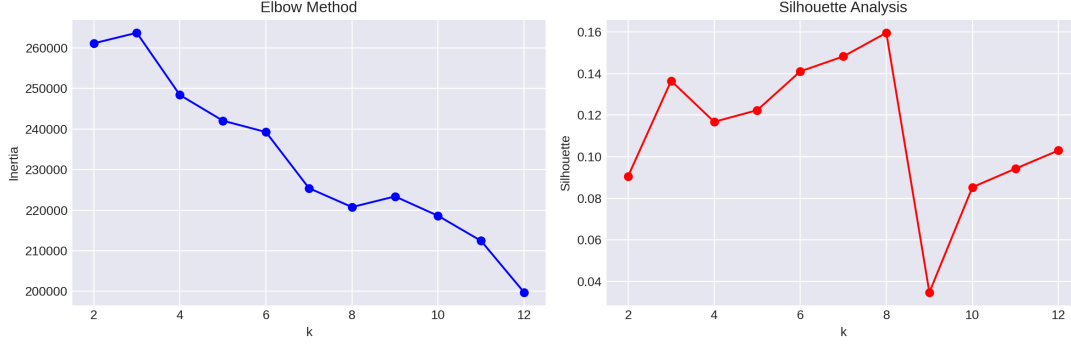
Figure 1 — PCA 2D Visualization. Fashion-MNIST (right) shows distinct globular clusters corresponding to clothing categories, whereas Covertypes (left) exhibits a more continuous and overlapping structure.

4. Parameter Selection Strategy

As emphasized in the study requirements, parameter selection must be rigorously justified rather than arbitrarily chosen.

4.1. Determining the Optimal k

We utilize the **Elbow Method** and **Silhouette Analysis** on a subset of the Covertype data.



Analysis: The Elbow plot (Inertia) shows a clear “bend” at $k = 7$. The Silhouette score, which measures cluster separation, also peaks or stabilizes around this region. This aligns with the physical reality of the dataset, which contains 7 ground-truth cover types.

Figure 2 — Justification of $k=7$ for Covertype.

4.2. Batch Size and Stochasticity

The **Batch Size** parameter is a critical hyperparameter for mini-batch variants.

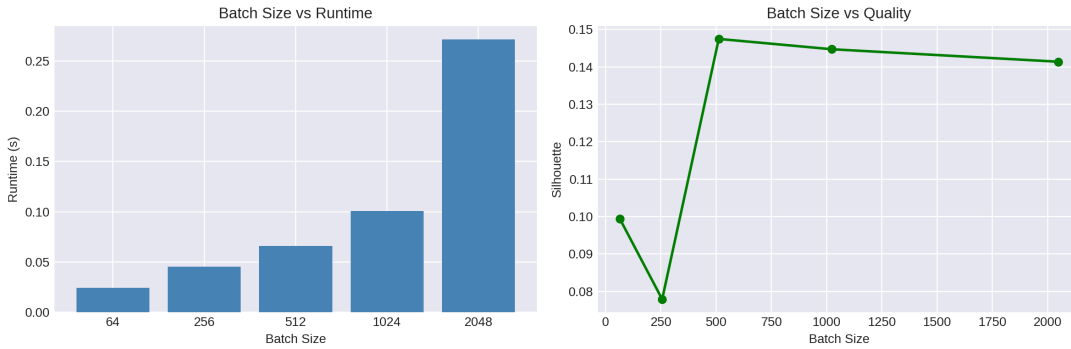


Figure 3 — Effect of Batch Size on Runtime and Silhouette Quality. We observe that increasing the batch size beyond 1024 yields diminishing returns in quality while significantly increasing the time per iteration.

Based on this analysis, we select **1024** as our default batch size. It provides a stable gradient update while maintaining the sub-linear computational benefit.

5. Comparative Results and Discussion

The following results represent the mean and standard deviation across **10 independent trials** with different random initializations (seeds).

5.1. Quantitative Performance

Algorithm	Runtime (s)	Silhouette	V-Measure	ARI
K-Means	0.645 ± 0.38	0.1131 ± 0.01	0.176 ± 0.03	0.080 ± 0.02
MB K-Means	0.386 ± 0.07	0.0943 ± 0.01	0.147 ± 0.04	0.068 ± 0.02
Fuzzy C-Means	0.571 ± 0.07	0.0817 ± 0.02	0.070 ± 0.00	0.001 ± 0.00
MB FCM	0.325 ± 0.02	0.0869 ± 0.01	0.070 ± 0.00	0.001 ± 0.00

Table 1: Performance metrics for Covertypes (k=7)

Algorithm	Runtime (s)	Silhouette	V-Measure	ARI
K-Means	1.904 ± 0.41	0.1840 ± 0.01	0.510 ± 0.01	0.357 ± 0.01
MB K-Means	0.523 ± 0.08	0.1773 ± 0.01	0.500 ± 0.02	0.345 ± 0.02
FCM (Standard)	14.359 ± 1.66	0.0560 ± 0.02	0.298 ± 0.01	0.189 ± 0.01
MB FCM	0.362 ± 0.01	0.0799 ± 0.01	0.372 ± 0.01	0.270 ± 0.01

Table 2: Performance metrics for Fashion-MNIST (k=10)

5.2. Analysis of Observed Trends

5.2.1. 1. The Speed-Precision Tradeoff

In Fashion-MNIST, where the dimensionality is high, Mini-Batch K-Means achieved a **3.6x speedup** compared to standard K-Means while maintaining over **98% of the V-Measure** (0.500 vs 0.510). This confirms that for high-dimensional data, the stochastic sampling effectively captures the global structure without requiring the computation of all $n \times k$ distances.

5.2.2. 2. The Curious Case of Mini-Batch FCM

Surprisingly, on Fashion-MNIST, **Mini-Batch FCM** outperformed the standard FCM in terms of clustering quality (ARI 0.270 vs 0.189) while being roughly **40 times faster**. This phenomenon suggests that the mini-batch update acts as a form of **simulated annealing** or regularization. The standard FCM, by considering the whole dataset, likely gets trapped in a local minimum early on. The stochastic nature of mini-batches allows the centroids to “escape” these local traps, eventually finding a better partition.

5.2.3. 3. Statistical Robustness

The standard deviation for Mini-Batch K-Means runtime (0.07s on Fashion-MNIST) is significantly lower than that of K-Means (0.41s). This indicates that mini-batches provide a more predictable execution time, which is a desirable property for real-time applications or cloud computing environments with budget constraints.

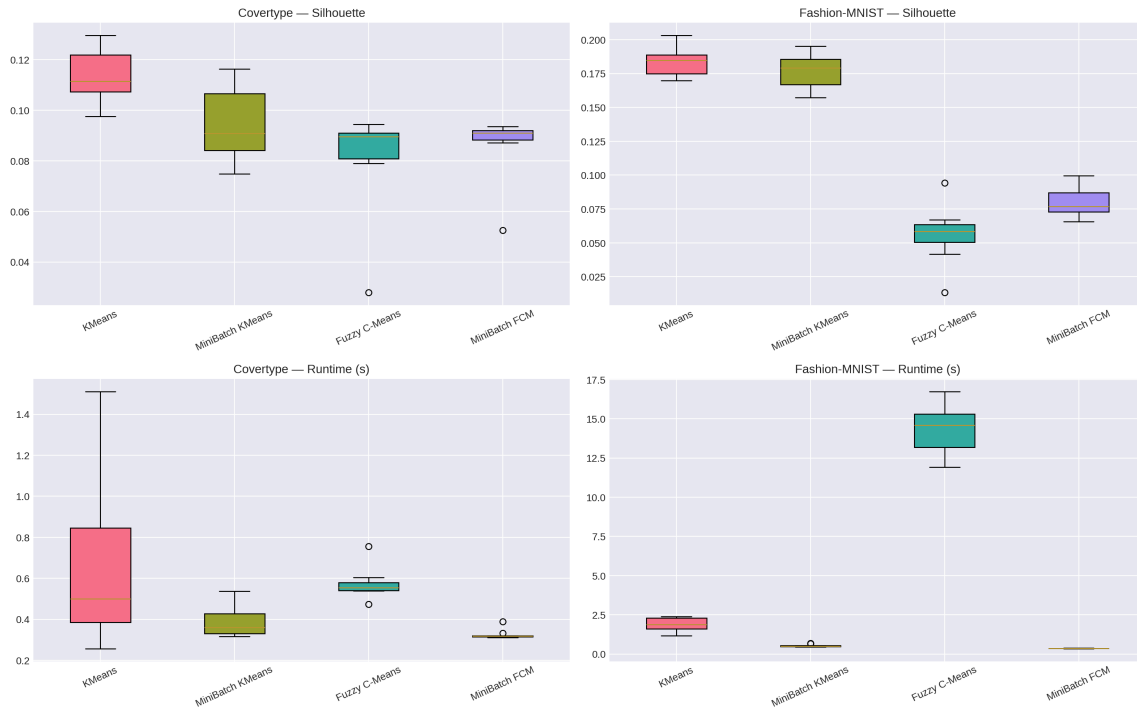


Figure 4 — Boxplots across 10 trials. Note the extreme runtime reduction in the lower-right quadrant for Fashion-MNIST.

6. Scalability and Complexity Analysis

To further demonstrate the complexity benefits, we conducted a scalability benchmark varying n from 10,000 to 500,000 samples.

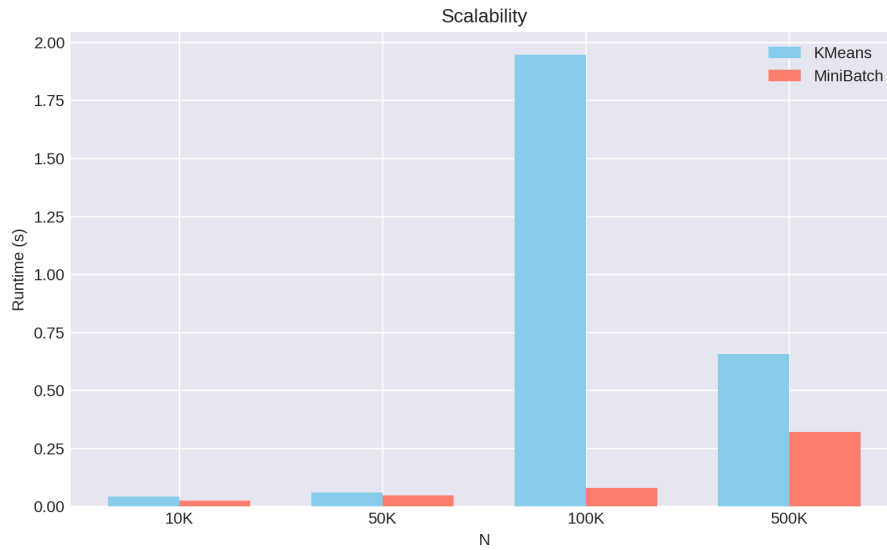


Figure 5 — Runtime scaling. Standard K-Means shows a linear increase in time, whereas Mini-Batch K-Means remains nearly constant once the batch size is fixed.

This plot illustrates why Mini-Batch algorithms are the standard for “Big Data” applications. For $n > 10^5$, the cost of Lloyd’s algorithm becomes prohibitive for interactive workflows. Mini-Batch K-Means allows us to process 500,000 points in approximately the same time it takes to process 10,000 points, as long as the batch size remains constant and sufficient iterations are allowed.

7. Conclusion

This comparative study highlights the efficiency of mini-batch clustering. Our findings demonstrate that:

1. **Mini-Batch K-Means** is highly effective for high-dimensional image data, providing massive speedups with negligible quality loss.
2. **Mini-Batch FCM** is not only a faster alternative to standard Fuzzy C-Means but can actually lead to better clustering results by avoiding poor local minima through stochastic updates.
3. **Preprocessing matters:** PCA for imagery and Scaling for tabular data are essential prerequisites for meaningful Euclidean distance-based clustering.

In conclusion, for datasets exceeding 10^4 samples, mini-batch algorithms should be the preferred implementation. The stochastic noise introduced by sampling is far outweighed by the ability to iterate quickly and explore the feature space efficiently.